

CS2601 Linear and Convex Optimization

Homework 10 Solution

Due: 2021.12.9

1. Consider the following problem,

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad & f(\mathbf{x}) = x_1^2 + x_2^2 \\ \text{s.t.} \quad & g_1(\mathbf{x}) = (x_1 - 1)^2 + (x_2 - 1)^2 - 1 \leq 0 \\ & g_2(\mathbf{x}) = (x_1 - 1)^2 + x_2^2 - 1 \leq 0 \end{aligned}$$

Write down the KKT conditions and find the optimal point $\mathbf{x}^*$ and the corresponding Lagrange multipliers.

Solution. The KKT conditions are

$$\begin{cases} 2x_1 + 2\mu_1(x_1 - 1) + 2\mu_2(x_1 - 1) = 0 \\ 2x_2 + 2\mu_1(x_2 - 1) + 2\mu_2x_2 = 0 \\ \mu_1 \geq 0 \\ \mu_2 \geq 0 \\ \mu_1[(x_1 - 1)^2 + (x_2 - 1)^2 - 1] = 0 \\ \mu_2[(x_1 - 1)^2 + x_2^2 - 1] = 0 \\ (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1 \\ (x_1 - 1)^2 + x_2^2 \leq 1 \end{cases}$$

Case I. Both g_1 and g_2 are active. Then

$$\begin{cases} (x_1 - 1)^2 + (x_2 - 1)^2 = 1 \\ (x_1 - 1)^2 + x_2^2 = 1 \end{cases}$$

so

$$x_2 = \frac{1}{2}, \quad x_1 = 1 \pm \frac{\sqrt{3}}{2}$$

Since $(1 - \frac{\sqrt{3}}{2}, \frac{1}{2})^T$ gives a smaller f value, $(1 + \frac{\sqrt{3}}{2}, \frac{1}{2})^T$ cannot be the optimal point, . Plugging $(1 - \frac{\sqrt{3}}{2}, \frac{1}{2})^T$

into the stationarity condition,

$$\begin{cases} 2 - \sqrt{3} - \sqrt{3}\mu_1 - \sqrt{3}\mu_2 = 0 \\ 1 - \mu_1 + \mu_2 = 0 \end{cases} \implies \begin{cases} \mu_1 = 1 - \frac{\sqrt{3}}{3} > 0 \\ \mu_2 = -\frac{\sqrt{3}}{3} < 0 \end{cases}$$

violating $\mu_2 \geq 0$.

Case II. g_1 is active, g_2 is inactive. Then $\mu_2 = 0$. Plugging into the stationarity condition,

$$\begin{cases} 2x_1 + 2\mu_1(x_1 - 1) = 0 \implies x_1 = \frac{\mu_1}{1+\mu_1} \\ 2x_2 + 2\mu_1(x_2 - 1) = 0 \implies x_2 = \frac{\mu_1}{1+\mu_1} \end{cases}$$

In particular $x_1 = x_2 \in [0, 1)$. Using $g_1(\mathbf{x}) = 0$,

$$x_1 = x_2 = 1 - \frac{\sqrt{2}}{2}, \quad \mu_1 = \frac{x_1}{1 - x_1} = \sqrt{2} - 1 > 0$$

We can check $g_2(\mathbf{x}) < 0$ indeed holds, as

$$(x_1 - 1)^2 + x_2^2 = 2 - \sqrt{2} < 1$$

Since the objective function is strictly convex, $\mathbf{x}^* = (1 - \frac{\sqrt{2}}{2}, 1 - \frac{\sqrt{2}}{2})^T$ is the unique global minimum. But let's check the other cases as well

Case III. g_2 is active, g_1 is inactive. Then $\mu_1 = 0$. Plugging into the stationarity condition,

$$\begin{cases} 2x_1 + 2\mu_2(x_1 - 1) = 0 \implies x_1 = \frac{\mu_2}{1+\mu_2} \\ 2x_2 + 2\mu_2x_2 = 0 \implies x_2(1 + \mu_2) = 0 \end{cases}$$

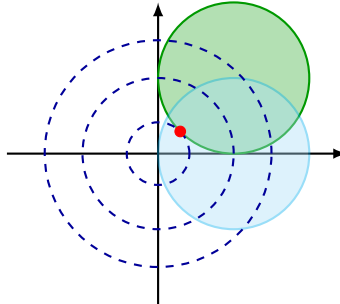
Since $\mu_2 \geq 0$, we have $x_2 = 0$. Plugging into $g_2(\mathbf{x}) = 0$,

$$x_1 = 0 \quad \text{or} \quad x_1 = 2$$

But both solutions violate $g_1(\mathbf{x}) < 0$, as

$$(x_1 - 1)^2 + (x_2 - 1)^2 = 2 > 1$$

Case IV. Both g_1 and g_2 are inactive. Then $\mu_1 = \mu_2 = 0$. Plugging into the stationarity condition, we obtain $x_1 = x_2 = 0$, violating both $g_1(\mathbf{x}) < 0$ and $g_2(\mathbf{x}) < 0$.



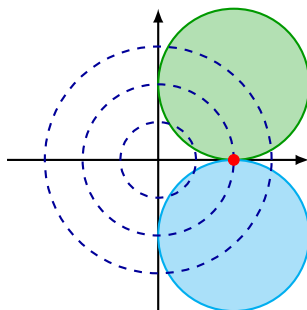
□

2. Consider the following problem

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad & f(\mathbf{x}) = x_1^2 + x_2^2 \\ \text{s.t.} \quad & (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1 \\ & (x_1 - 1)^2 + (x_2 + 1)^2 \leq 1 \end{aligned}$$

- (a). Sketch the feasible set and the level set of the objective function. Find the optimal point $\mathbf{x}^*$ and the optimal value f^* .
- (b). Write down the KKT conditions. Do there exist Lagrange multipliers that satisfy the KKT conditions? Is $\mathbf{x}^*$ regular?

Solution. (a). The only feasible point is $\mathbf{x}^* = (1, 0)^T$, shown by the red dot. The dashed circles are the level sets of the objective function. The optimal value is $f^* = f(\mathbf{x}^*) = 1$.



(b). The KKT conditions are

$$\begin{cases} \mu_1 \geq 0 \\ \mu_2 \geq 0 \\ 2x_1 + 2\mu_1(x_1 - 1) + 2\mu_2(x_1 - 1) = 0 \\ 2x_2 + 2\mu_1(x_2 - 1) + 2\mu_2(x_2 + 1) = 0 \\ \mu_1[(x_1 - 1)^2 + (x_2 - 1)^2 - 1] = 0 \\ \mu_2[(x_1 - 1)^2 + (x_2 + 1)^2 - 1] = 0 \\ (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1 \\ (x_1 - 1)^2 + (x_2 + 1)^2 \leq 1 \end{cases}$$

In particular, the stationarity condition requires

$$2x_1 + 2\mu_1(x_1 - 1) + 2\mu_2(x_1 - 1) = 0$$

which leads to a contradiction $2 = 0$ at $\mathbf{x}^* = (1, 0)$. So no Lagrange multipliers satisfy the KKT condition at $\mathbf{x}^*$. The gradients of the constraints at $\mathbf{x}^*$ are

$$(0, -1)^T, \quad (0, 1)^T$$

which are linearly dependent, so $\mathbf{x}^*$ is not regular.

□

3. Consider the following problem

$$\begin{aligned} \min \quad & (x_1 - \frac{9}{4})^2 + (x_2 - 2)^2 \\ \text{s.t.} \quad & -x_1^2 + x_2 \geq 0 \\ & x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0 \end{aligned}$$

For each of the following points, determine whether it is an optimal solution to the above problem and show your arguments,

$$\mathbf{x}^{(1)} = \begin{bmatrix} \frac{9}{4} \\ 2 \end{bmatrix}, \mathbf{x}^{(2)} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \mathbf{x}^{(3)} = \begin{bmatrix} \frac{3}{2} \\ \frac{9}{4} \end{bmatrix}$$

Hint: Try if you can find Lagrange multipliers satisfying the KKT conditions. Note you can easily check which constraints are active.

Solution. The problem can be written as:

$$\begin{aligned} \min \quad & f(\mathbf{x}) = (x_1 - \frac{9}{4})^2 + (x_2 - 2)^2 \\ \text{s.t.} \quad & g_1(\mathbf{x}) = x_1^2 - x_2 \leq 0 \\ & g_2(\mathbf{x}) = x_1 + x_2 - 6 \leq 0 \\ & g_3(\mathbf{x}) = -x_1 \leq 0 \\ & g_4(\mathbf{x}) = -x_2 \leq 0 \end{aligned}$$

As both the objective function and the inequality constraints are convex, we only need to see if the given points satisfy the KKT condition.

- $\mathbf{x}^{(1)}$: $\mathbf{x}^{(1)}$ is not a feasible point, as

$$g_1(\mathbf{x}_1^{(1)}) = \frac{81}{16} - 2 > 0.$$

- $\mathbf{x}^{(2)}$: $\mathbf{x}^{(2)}$ is a feasible point and only $g_3(\mathbf{x}) \leq 0$ is active. The (reduced) KKT conditions are

$$\begin{cases} 2(x_1 - \frac{9}{4}) - \mu_3 = 0 \\ 2(x_2 - 2) = 0 \\ \mu_3 \geq 0 \end{cases}$$

From the first equation we obtain $\mu_3 = 2(x_1^{(2)} - \frac{9}{4}) = -\frac{9}{2}$, violating $\mu_3 \geq 0$. There are no Lagrange multipliers satisfying the KKT conditions at $\mathbf{x}^{(2)}$. As $\nabla g_3(\mathbf{x}^{(2)}) = (-1, 0)^T \neq \mathbf{0}$, $\mathbf{x}^{(2)}$ is regular and the KKT conditions are necessary for optimality. Thus $\mathbf{x}^{(2)}$ cannot be optimal.

- $\mathbf{x}^{(3)}$: $\mathbf{x}^{(3)}$ is a feasible point and only $g_1(\mathbf{x})$ is active. The (reduced) KKT conditions are

$$\begin{cases} 2(x_1 - \frac{9}{4}) + 2\mu_1 x_1 = 0 \\ 2(x_2 - 2) - \mu_1 = 0 \\ \mu_1 \geq 0 \end{cases}$$

which are satisfied by $\mu_1 = \frac{1}{2}$. Since the problem is convex and the KKT conditions hold, $\mathbf{x}^{(3)}$ must be optimal.

□

4. Consider the following optimization problem,

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \|\mathbf{x} - \mathbf{z}\|^2 \\ \text{s.t.} \quad & \mathbf{y}^T \mathbf{x} = 0 \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

where $\mathbf{z} \in \mathbb{R}^n$ and $\mathbf{y} \in \{1, -1\}^n$ are constants.

- (a). Show that there exists a $\lambda \in \mathbb{R}$ s.t. the solution $\mathbf{x}^*$ takes the form

$$x_i^* = (z_i - \lambda y_i)^+, \quad i = 1, 2, \dots, n$$

where λ satisfies

$$\sum_{i=1}^n y_i (z_i - \lambda y_i)^+ = 0$$

You can assume the KKT conditions hold even at a critical optimal point.

Remark. The KKT conditions hold even at critical optimal points because the constraint functions here are all affine. In our discussion of the Lagrange condition for equality constrained convex problems, we assumed the matrix $\mathbf{A}$ has full row rank for simplicity. When $\mathbf{A}$ has dependent rows, we can eliminate the redundant ones and set the corresponding Lagrange multipliers to zero.

- (b). Let $I_+ = \{i : y_i = 1\}$ and $I_- = \{i : y_i = -1\}$. Relabel elements in $\{z_i : i \in I_+\}$ as $u_1 \leq u_2 \leq \dots \leq u_p$, and those of $\{-z_i : i \in I_-\}$ as $w_1 \leq w_2 \leq \dots \leq w_m$. We also use the convention $u_0 = w_0 = -\infty$ and $u_{p+1} = w_{m+1} = +\infty$. Then λ satisfies

$$\sum_{i=1}^p (u_i - \lambda)^+ = \sum_{j=1}^m (\lambda - w_j)^+$$

If $u_k \leq \lambda \leq u_{k+1}$ and $w_\ell \leq \lambda \leq w_{\ell+1}$, then

$$\lambda = \frac{\sum_{i=k+1}^p u_i + \sum_{j=1}^{\ell} w_j}{p - k + \ell}$$

If λ indeed satisfies $u_k \leq \lambda \leq u_{k+1}$ and $w_\ell \leq \lambda \leq w_{\ell+1}$, then it is the solution. Once you have λ , you can find $\mathbf{x}^*$. Algorithm 1 shows one way to solve the problem in $O(n \log n)$ time. Write a python

Algorithm 1

```
1: if  $p = 0$  or  $m = 0$  then
2:   return  $\mathbf{x} \leftarrow \mathbf{0}$ 
3: end if
4:  $\mathbf{u} \leftarrow \text{sort}(\mathbf{z}[\mathbf{y} > 0])$ 
5:  $\mathbf{w} \leftarrow \text{sort}(-\mathbf{z}[\mathbf{y} < 0])$ 
6:  $U_p \leftarrow 0$ ;  $U_{k-1} \leftarrow U_k + u_k$  for  $k = p, p-1, \dots, 1$   $\triangleright U_k = \sum_{i=k+1}^p u_i$ 
7:  $W_0 \leftarrow 0$ ;  $W_\ell \leftarrow W_{\ell-1} + w_\ell$  for  $\ell = 1, \dots, m$   $\triangleright W_\ell = \sum_{j=1}^\ell w_j$ 
8:  $k \leftarrow 0, \ell \leftarrow 0$ 
9: while  $k \leq p$  and  $\ell \leq m$  do
10:    $\lambda \leftarrow (U_k + W_\ell) / (p - k + \ell)$ 
11:   if  $u_k \leq \lambda \leq u_{k+1}$  and  $w_\ell \leq \lambda \leq w_{\ell+1}$  then  $\triangleright u_0 = w_0 = -\infty$  and  $u_{p+1} = w_{m+1} = +\infty$ 
12:     break
13:   end if
14:   if  $u_{k+1} < w_{\ell+1}$  then
15:      $k \leftarrow k + 1$ 
16:   else
17:      $\ell \leftarrow \ell + 1$ 
18:   end if
19: end while
20: return  $\mathbf{x} \leftarrow (\mathbf{z} - \lambda \mathbf{y})^+$ 
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function that takes an arbitrary $\mathbf{z}$ and $\mathbf{y}$ as input and outputs the optimal solution $\mathbf{x}^*$. Show the result for $\mathbf{z} = (1, 2, 1)^T$ and $\mathbf{y} = (1, 1, -1)^T$. Note that you can find λ graphically for this simple case.

Solution. (a). The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \lambda, \boldsymbol{\mu}) = \frac{1}{2} \|\mathbf{x} - \mathbf{z}\|^2 + \lambda \mathbf{y}^T \mathbf{x} - \boldsymbol{\mu}^T \mathbf{x}$$

By the stationarity condition,

$$\mathbf{x} - \mathbf{z} + \lambda \mathbf{y} - \boldsymbol{\mu} = \mathbf{0} \implies \mathbf{x} = \mathbf{z} - \lambda \mathbf{y} + \boldsymbol{\mu}$$

If $x_i > 0$, then $\mu_i = 0$ and

$$x_i = z_i - \lambda y_i > 0$$

If $x_i = 0$, then

$$z_i - \lambda y_i = -\mu_i \leq 0$$

Thus

$$x_i = (z_i - \lambda y_i)^+$$

By the equality constraint,

$$\sum_{i=1}^n y_i (z_i - \lambda y_i)^+ = 0$$

(b).

$$\mathbf{x}^* = \left(\frac{1}{3}, \frac{4}{3}, \frac{5}{3}\right)^T \approx (0.33333333, 1.33333333, 1.66666667)^T$$

□